Concept Memory: 3-Mode Experiment

Baseline vs Concept vs Hybrid

gpt-5-nano | 4 runs | 2 passes | Competition Math

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Setup

3 modes, each with 2 passes (pass 1 = initial, pass 2 = retry on failures):

Mode	Pass 1	Pass 2 (retry)
Baseline	No concepts	No concepts
Concept	With concepts	With concepts
Hybrid	No concepts	Concepts on retry only

- 4 independent runs per mode, gpt-5-nano, `n=1`
- 200 Level-5 competition math problems
- 837 concepts extracted from 420 build solutions
- Selection coverage: 196/200 (98%)

Results: Per-Run Scores

Mode	Run 1	Run 2	Run 3	Run 4	Mean
Baseline p1	178	176	173	171	174.5
Concept p1	182	175	183	175	178.8
Hybrid p1	174	171	173	171	172.2
Baseline best	184	187	188	185	186.0
Concept best	187	187	184	187	186.2
Hybrid best	188	190	187	189	188.5

Results: Summary

Metric	Baseline	Concept	Hybrid
Mean p1	174.5	**178.8** (+4.3)	172.2
Mean best	186.0	186.2	**188.5** (+2.5)
Retry gain	+11.5	+7.5	**+16.2**
Oracle (4-run)	193	193	**194**

- **Concept** boosts first-attempt accuracy by **+4.3**
- **Hybrid** has strongest retry: fresh attempt + concept hint = **+16.2** gain
- Concept retry gain is *lowest* (7.5) — same hint on both passes has diminishing returns
- Oracle union of all three modes: **195/200**

Why Hybrid Wins on Retry

Hybrid = best of both worlds:

1. **Pass 1 without hints** avoids misleading concepts on easy problems
2. **Pass 2 adds concepts only to failures** — problems that genuinely need help
3. Error feedback + concept hint = strongest retry signal

Concept mode uses hints on both passes, so pass 2 repeats the same hint with diminishing returns (+7.5 vs hybrid's +16.2)

Baseline retries without any new information — only error feedback drives improvement (+11.5)

Analysis

What do the concepts look like? When do they help vs hurt?

Case: Concept Helps (cmath_2920)

Problem: Two intersecting chords in a circle, radius 5, BC=6, AD bisected by BC. Find sine of central angle of minor arc AB as $\frac{m}{n}$, compute mn.

Hint given:

```
- concept: Center-based Relative Coordinate System
  description: Use coordinates relative to center to simplify symmetry.
  cues:
    - dx = i-4, dy = j-4
  implementation:
    - classify point by (|dx|,|dy|) to identify orbit under D4
- concept: Distance from a Point to a Line
  description: Perpendicular distance from point to line = triangle height.
  cues:
    - Line BC: x - 11y + 197 = 0
    - dist(A, BC) = height
  implementation:
    - distance = |p - 11q + 197| / √122
```

Useful geometric framing. Baseline fails; concept mode solves it.

Case: Concept Hurts (cmath_5298)

Problem: Cars on a highway obey safety rule: distance = one car length per 15 km/h. Photoelectric eye counts cars per hour. Maximize M/N where N=number of cars, M=speed in km/h.

Baseline solves 4/4 runs. Concept solves 1/4.

Hint given:

```
- concept: Speed Unit Interpretation
  description: speed = distance/time; 160mi ÷ 5hr = 32mph
- concept: Interval Width
  description: width of [a,b] is b - a
- concept: Floor-Ceiling Interval Bound for Integer Counting
  description: count integers in [a,b] = ⌊b⌋ - ⌈a⌉ + 1
  implementation:
    - n = 1+23k, enforce -9999 ≤ n ≤ -1000
    - k_max = floor((-1001)/23), n = 1+23·k_max = -1011
```

Leakage: "160 miles / 5 hours = 32 mph" and "-9999 $\leq$ n $\leq$ -1000" are values from the **source** problem, not this one. Injects contradictory numbers that confuse the model.

Case: Concept Hurts (cmath_2931)

Problem: Rectangle ABCD ($12\sqrt{3} \times 13\sqrt{3}$), cut triangle ABP, fold into pyramid. Find volume.

Baseline 4/4, Concept 2/4. Hints: *Scalar Triple Product*, *Triangle Area Formula*

```
- concept: Triangle Area Formula
  implementation:
    - ACM: base CM = 2, height = 8 → area = 8
    - ACN: base CN = 4, height = 4 → area = 8
```

Again, **source-specific values** (CM=2, height=8, CN=4) leak into the hint. These numbers belong to a different problem entirely.

Failure Taxonomy

Failure Mode	Effect	Frequency
Misleading	Suggests wrong approach	~2-3 per run
Too generic	Obvious hint, wastes tokens	~3-4 per run
Leakage	Source-specific values in hint	~1-2 per run

Net effect per run: **~4 wins, ~2 harms** from concepts
(+4.3 p1 mean = genuine positive signal despite noise)

Key observation

Concepts help more than they hurt on math — the 837-concept pool with 98% coverage provides enough relevant matches. The remaining harms come from selection quality, not pool size.

Takeaways

Summary

- **Concept mode**: best for p1 accuracy (**+4.3**)
- **Hybrid mode**: best for final score (**+2.5 best**, +16.2 retry gain)
- **Baseline**: weakest retry (+11.5), but cleanest p1 signal
- Oracle across all modes: **195/200** (only 5 never solved)

Concepts provide genuine signal on competition math when:

- Large concept pool (837 from 420 solutions)
- High selection coverage (98%)
- Domain has transferable techniques (algebra, geometry, number theory)